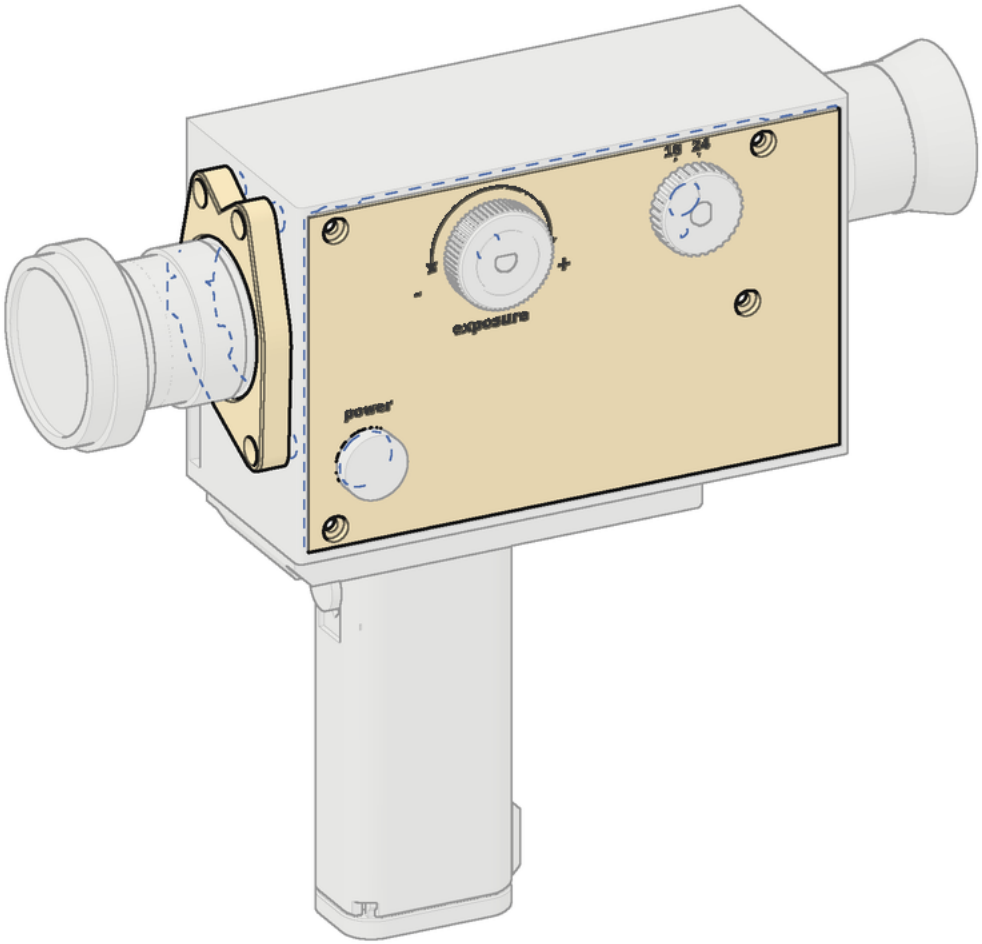
D2 Assembly and wiring with concealed vents

Vertical Pi candidate | 158 x 63.6 x 100 mm core | 9 printed parts | revised review



A sequence to test

Pictures expose fit, insertion and service dependencies.

Amber: new part.
Grey: already present.
Blue: proposed motion.
Dashed: hidden part.

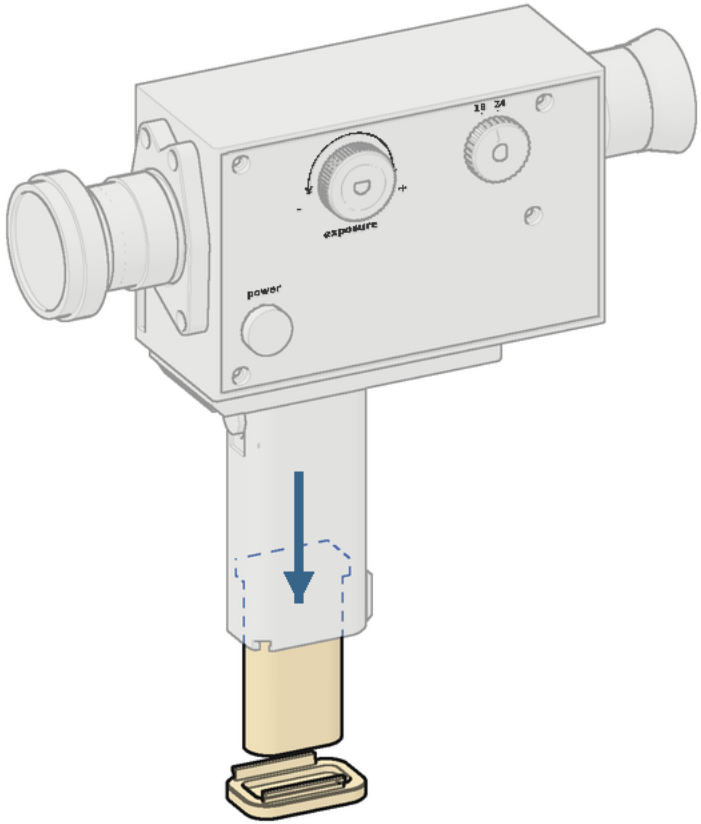
The outer plenum is integral to the tub.

No physical build is qualified.

STOP All 56 inherited physical gates and M1 remain open. Do not fabricate or energize from this review.

00 Disconnect before opening

The J2 button requests power control. It does not isolate the battery.



Four prerequisites

1 Stop recording; clean shutdown.

2 Remove every external supply.

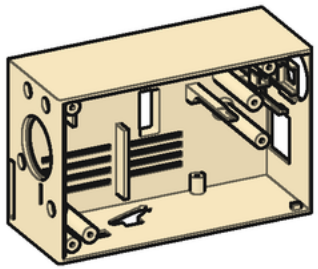
3 Withdraw pack; part XT30 by housings. Protect contacts.

4 Only then touch internal connectors. Use ESD protection. Charge the pack outside the body.

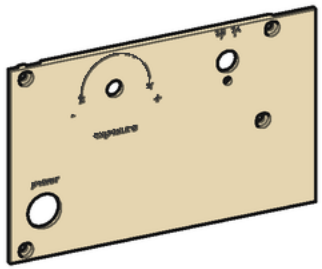
STOP XT30 must be reachable with body closed and retention fitted. Fixed tied loops cannot unfold for service.

01 Count the printed parts

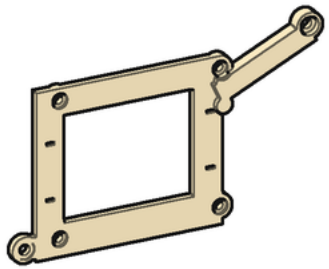
Nine compact production parts, each x1. Gauges and plenum / structural coupons are separate test pieces.



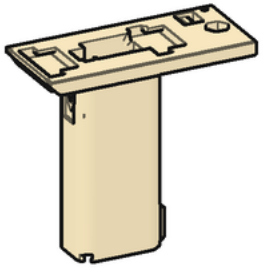
1 Tub



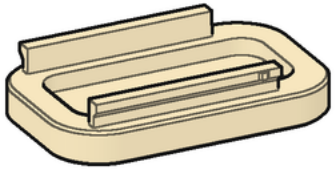
2 Panel



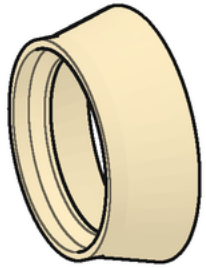
3 Carrier



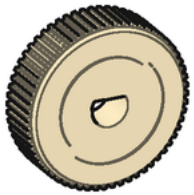
4 Base / grip



5 Battery cap



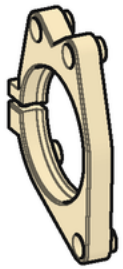
6 Eyecup



7 Exposure knob



8 18 / 24 knob

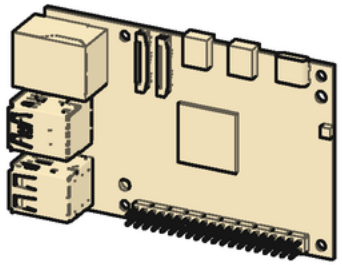


9 Lens collar

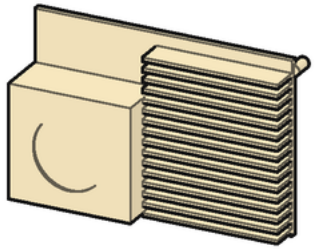
Same 9 prints; no vent covers or added fasteners. Panel is plain 2.8 mm. Purchased parts are unchanged.

01b Identify the purchased assemblies

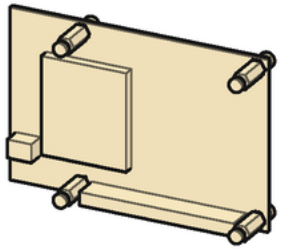
CAD envelopes. Match actual purchased identities and revisions; each group is one installed assembly.



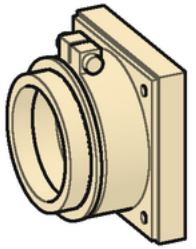
A Raspberry Pi 5



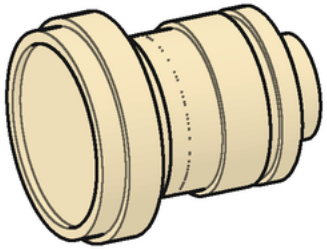
B Active Cooler envelope



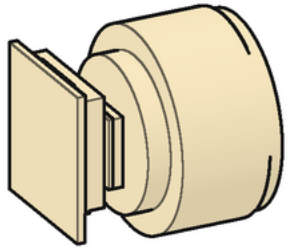
C X1203 + kit hardware



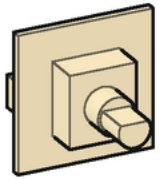
D GS camera + C-CS



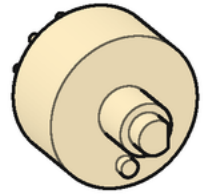
E Kowa LM6HC lens



F EVF kit (flex not shown)



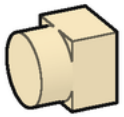
G QT encoder



H 18 / 24 switch



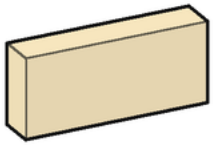
I J2 power switch



J Run button



K Protected pack + XT30



L Princeton USB SSD

Also identify microSD, camera FPC, HDMI heads + ribbon, EVF feed, QT / switch / run / J2 leads, fuse and insulation.

02 Sort hardware and tools

Counts are installed quantities. Purchased spares are not extra installed hardware.

A

7 x PT 3.0 x 12 PH1

Panel 4 | carrier 2 | independent grip lock 1

B

3 x M3 x 10 + washers

Collar anchors into 3 short M3 inserts

C

1 x M3 x 16 + washer

Collar pinch into 1 long M3 insert

D

4 x M2.5 x 5 retained

Upper kit screws; retain 4 original standoffs

E

4 x M2.5 x 8 replacements

Lower screws through carrier; bag 4 old M2.5 x 5

F

2 x supplied cooler push pins

Included in cooler kit; proxy feet are not fasteners

G

1 x tripod nut

Capture in grip before joining the tub

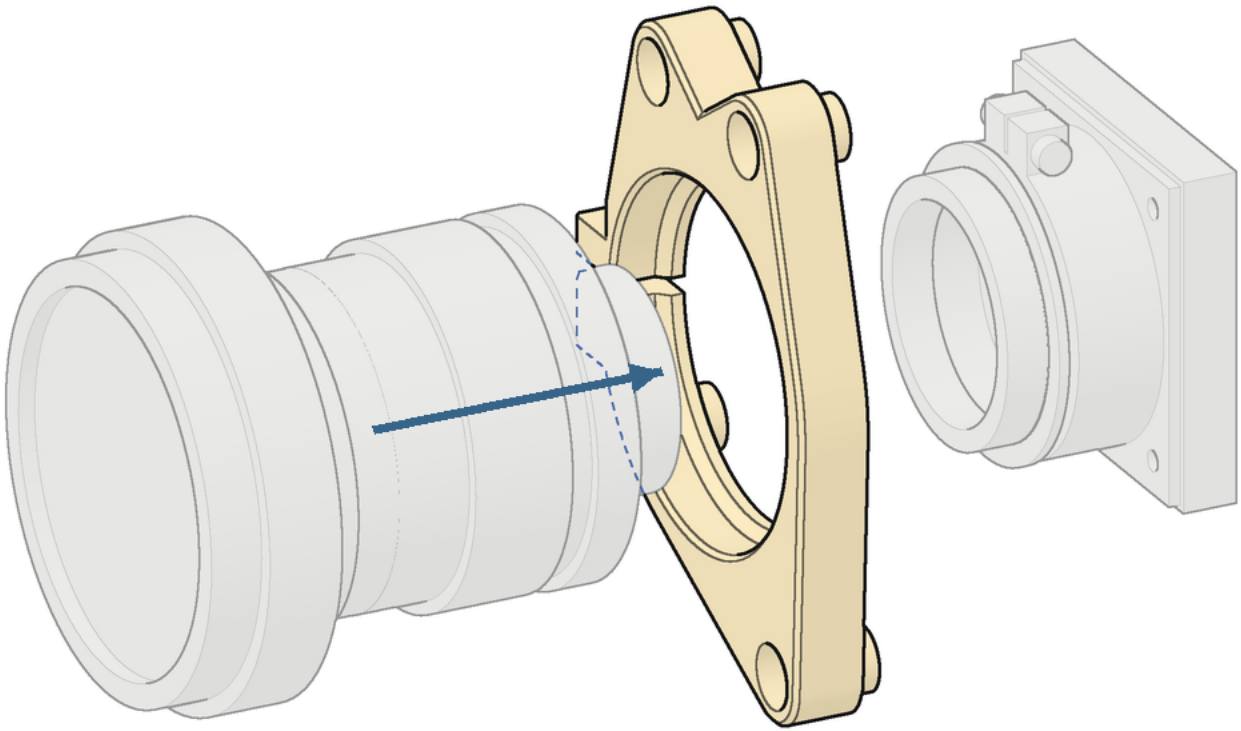
TOOLS

T1 PH1 + torque driver
T2 Iron + correct insert tip
T3 Solder / flux / wick + heat gun
T4 Stripper / cutter
T5 Meter, thermocouple, calipers
T6 Fine tweezers + smooth pliers
T7 Nut tools, saw, file, soft vise
T8 ESD mat + wrist strap
T9 Panel prop, shims, rule, gauge
T10 Bench supply, Pi PSU, charger
T11 Scope, USB meter, DC load
T12 Insert-repair pin vise / drills

STOP Torque and insert process need revised coupons. No threadlocker on ASA. Power-nut tool fit is unmeasured.

03 Prepare optics and prints

Bench preparation before electronics go inside



Support the load

Support camera and lens separately. Never react lens torque through PCB or cover.

Measure back focus and support band. Keep adapter / BFAR witness marks.

Bag tripod block after setup. Install inserts before electronics.

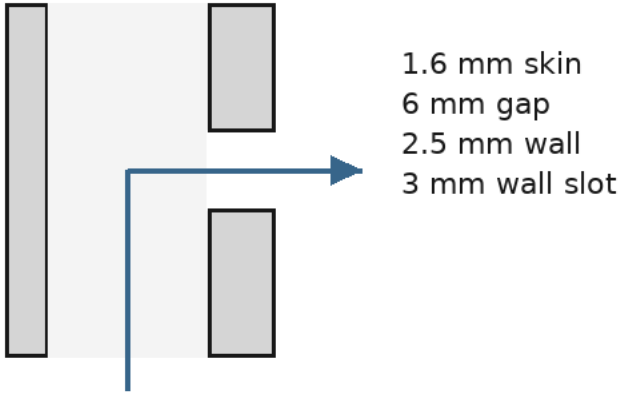
STOP Review the outer plenum on 03v / 03w plus collar and horizontal-bores supports. Measured optics and M1 remain open.

03v Inspect the concealed outer plenum

Representative sections. Blue arrows show proposed cold/hot paths, not measured airflow. Posts omitted for clarity.

COLD BAY / INTAKE

EXTERIOR CAVITY



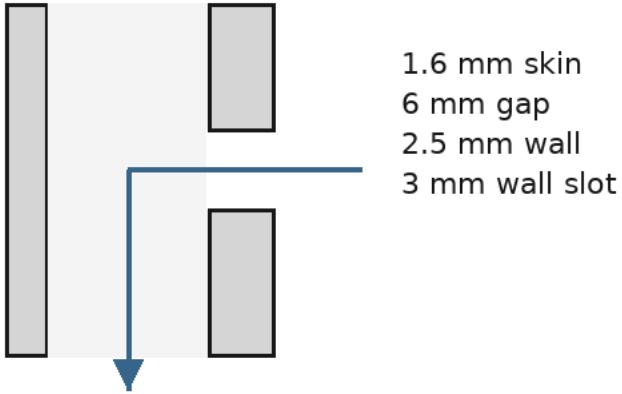
Underside mouth opens into the gap

Rear + underside perimeter mouths feed the cold bay. Four internal wall slots have 420 mm2 nominal total area.

Gross mouth areas: rear 420 mm2; underside 426 mm2.

HOT BAY / EXHAUST

EXTERIOR CAVITY



Underside mouth opens into the gap

Four internal wall slots feed the hot bay: 510 mm2 nominal total area. Air is intended to leave at front + underside perimeter mouths.

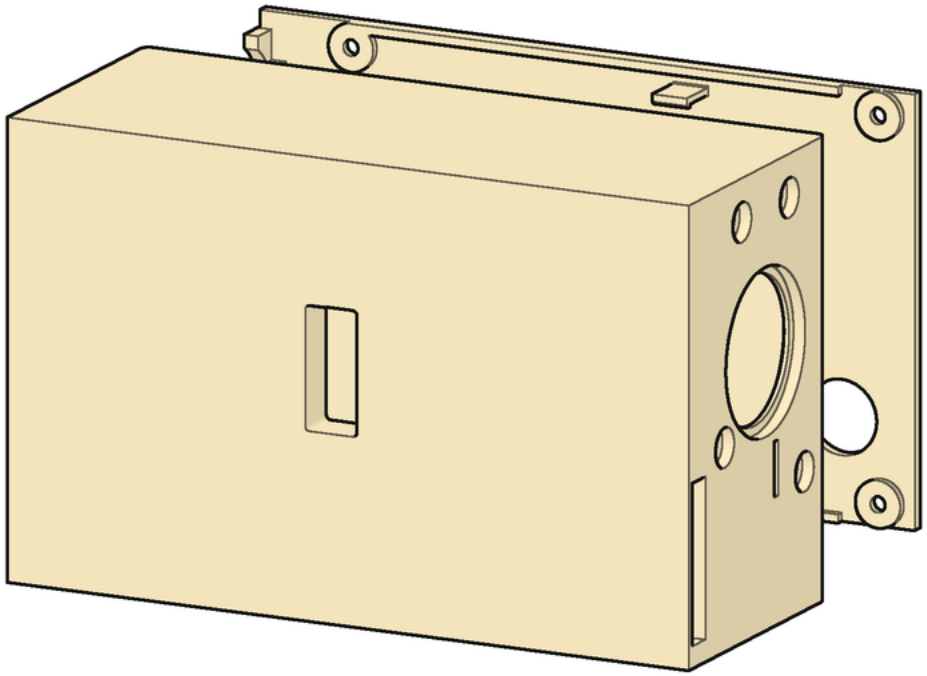
Gross mouth areas: front 210 mm2; underside 324 mm2.

Side skin 1.6 mm + gap 6 mm + original wall 2.5 mm. Cavity datum stays Y -25.5. Broad-side grille is concealed; perimeter mouths remain.

STOP CSI is an explicit service aperture. Direct rear/front views expose mouths. No sealed-flow, dust or water rating.

03w Print and inspect the outer plenum

Do not fit electronics or inserts until the full air path can be inspected and cleaned.



PRINT HOLD: the front mouth removes edge support at Z5..40. The 10.7 mm gap is not a supported bridge; toolpaths may need about 35 mm bridging, depending on raster direction.

Old panel grooves were blind, with 0.2 mm floors. The revised source specifies a plain 2.8 mm skin.

PRINT AND INSPECT

1 Current tub STL is -Y-face down. Inspect the skin, posts, chamber roof and every first bridge in the slicer.

2 Posts: diameter 1.6 mm, nominal grid 8 mm. Qualify the 48 x 32 grid and 110 x 100 port/CSI crops in the slicer and on the bench; both are geometry-only.

3 No trapped support. If slicing or either coupon fails, add local support webs and repeat the airflow-section checks before any new build trial.

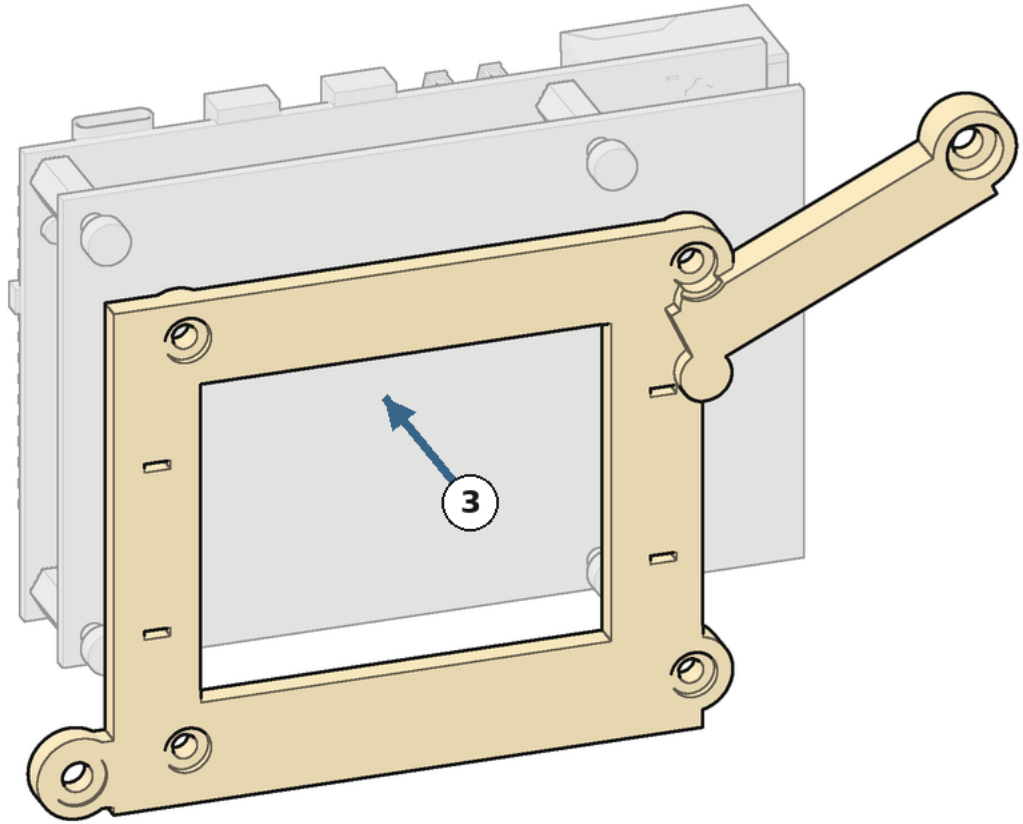
4 Open-edge coupons cannot prove closed-body cleanout. Inspect both actual bays through their mouths using light/mirror or a suitable scope; remove all debris.

5 Check four perimeter mouths and all eight inner-wall slots. If any chamber region cannot be inspected or cleared, stop; do not seal debris inside.

STOP Blocked slots, roof sag, loose posts or trapped support are rejects. This plenum has no slicer, bridge, cleanout or physical print pass.

04 Build one removable stack

Prepare J2, battery-pad and header wiring while both board faces remain accessible.



Keep kit spacing

Fit cooler with its 2 push pins and native fan lead once.

Retain upper 4 x M2.5 x 5 + original standoffs.

Lower 4 x M2.5 x 8 pass through carrier. Check thread depth before tightening.

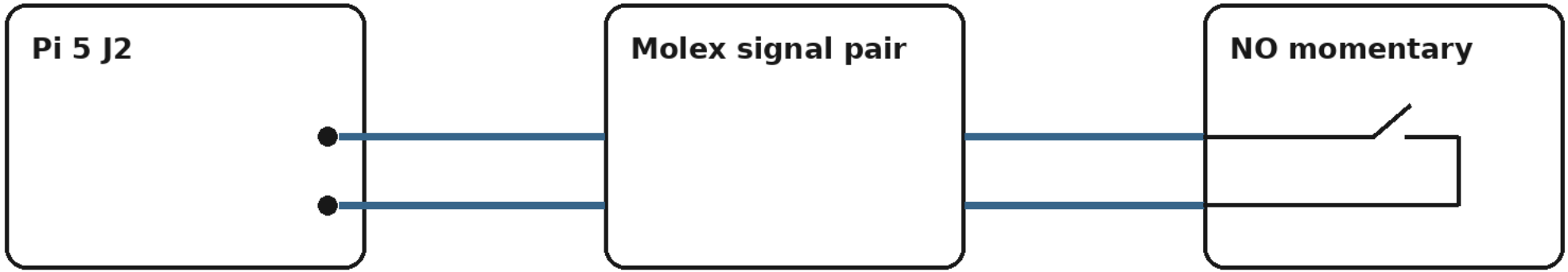
3.4 mm engagement is nominal only.

SOURCE / PROXY HOLD: registered Pi intersects the illustrative cooler perimeter / pin model (79.90 mm3). Measure and register the actual cooler outline before assembly.

STOP No battery. No assumed pad location. Do not change pogo geometry or repeatedly unclip the cooler.

05 Wire the detachable power button

Electrical schematic, not a measured pad-location drawing



Bridge only the two J2 pads with a normally-open momentary switch.

Identify J2 on the actual Pi: between RTC battery connector and board edge. Insulate and strain-relieve all four solder joints. Meter open / pressed continuity before power.

One 2181120201 + one 2181110202 pair. Both wires are black: trace continuity. Label PWR separately from FPS; their shared key does not prevent cross-connection.

STOP J2 is signal-only, not GPIO or battery isolation. Finished cuts, pad reach, wire exits and nut access remain gates.

06 Prepare control wiring

Physical J8 pin numbers below. This is a pin map, not a rotated-board orientation drawing.

QT encoder

Pin 1 3V3 -> SH 2
Pin 3 SDA -> SH 3
Pin 5 SCL -> SH 4
Pin 6 GND -> SH 1
Bridge A0: address 0x37

18 / 24

Pin 33 GPIO13
Pin 34 GND
CK1044 common A / throw 2
Closed / low = 24 fps
Open / high = 18 fps

Run button

Pin 37 GPIO26
Pin 39 GND

Normally open
Pressed / low = run

Match physical pin 1 to the real board silkscreen / official pinout before plugging.

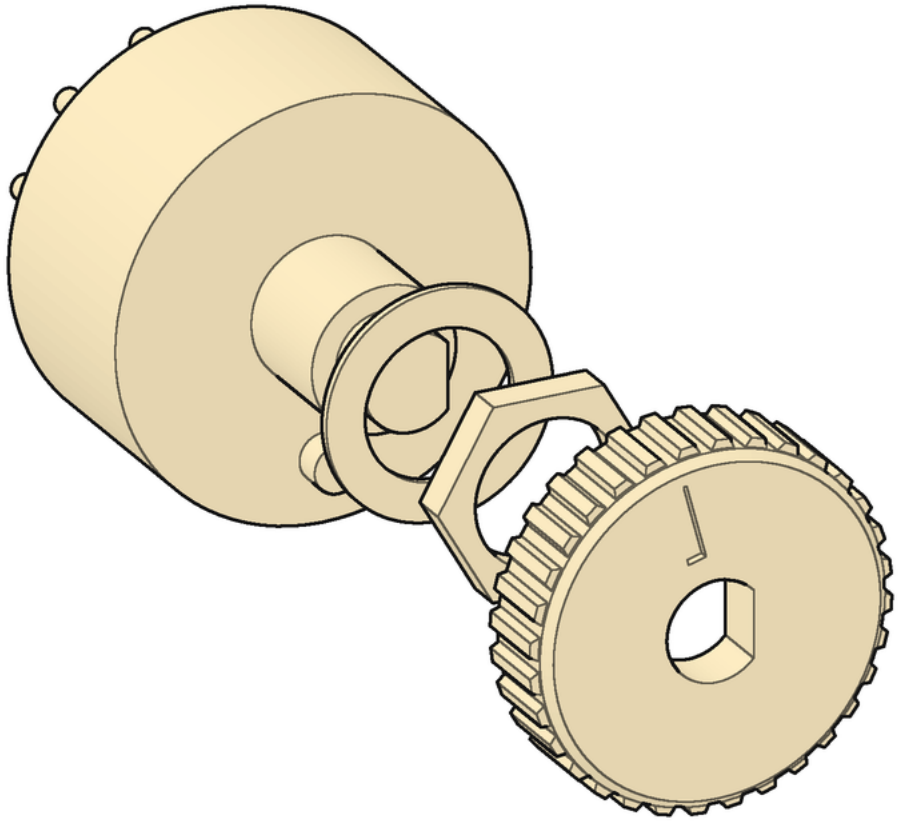
QT target: 250 +/- 10 mm, two source leads, four staggered insulated splices. Meter by SH pin number; never match wire colours alone. Place splice sleeves on a straight, accessible run.

The panel carries encoder and J2. Support the FPS switch on its tub rest; keep its complete pair mated during carrier service. Separate that pair only on the bench.

STOP Header plugs and every wire tail must clear the stack, HDMI, fan and carrier motion.

06b Set the real frame-rate switch

Conditional Lorlin CK1044. Two adjacent detents are 30 degrees apart.



Three distinct pieces

Nut. Mounting washer. Stop / tang washer.

Stop washer shape is not measured and is not drawn. Confirm supplied hardware.

Set two adjacent positions. Meter A / 2: open = 18; closed = 24.

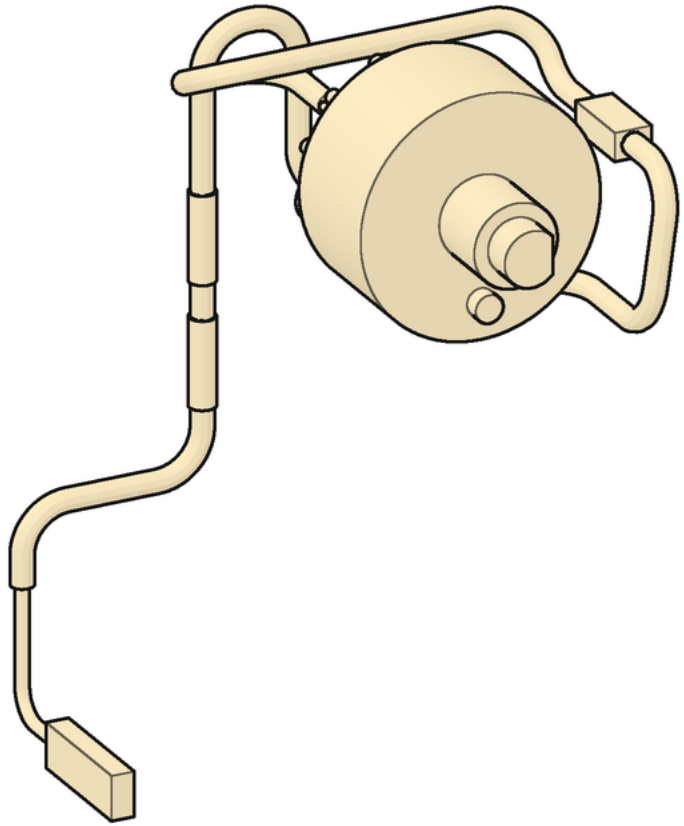
Preserve the stop setting when removing knob, nut and panel.

Measure the actual stack before cutting. Proposed shaft length 11.8 mm from mounting face is a fit study; 6.35 mm D shaft / 5.5 mm flat needs a real knob coupon.

STOP Nut / mounting-washer shapes are estimated. Actual stop stack, pin clocking, shaft cut and tool access remain gates.

06c Keep the full FPS harness together

Two donor female GPIO heads + two splices + both factory Molex housings + A / throw 2.



Identify both black wires

FPS-only CAD group clears; other looms remain gated.

Male 2181110202: 225 +/- 6 mm exposed stock.

Female 2181120201: 150 +/- 6 mm exposed stock.

200 mm main axis includes donor stubs / splices.

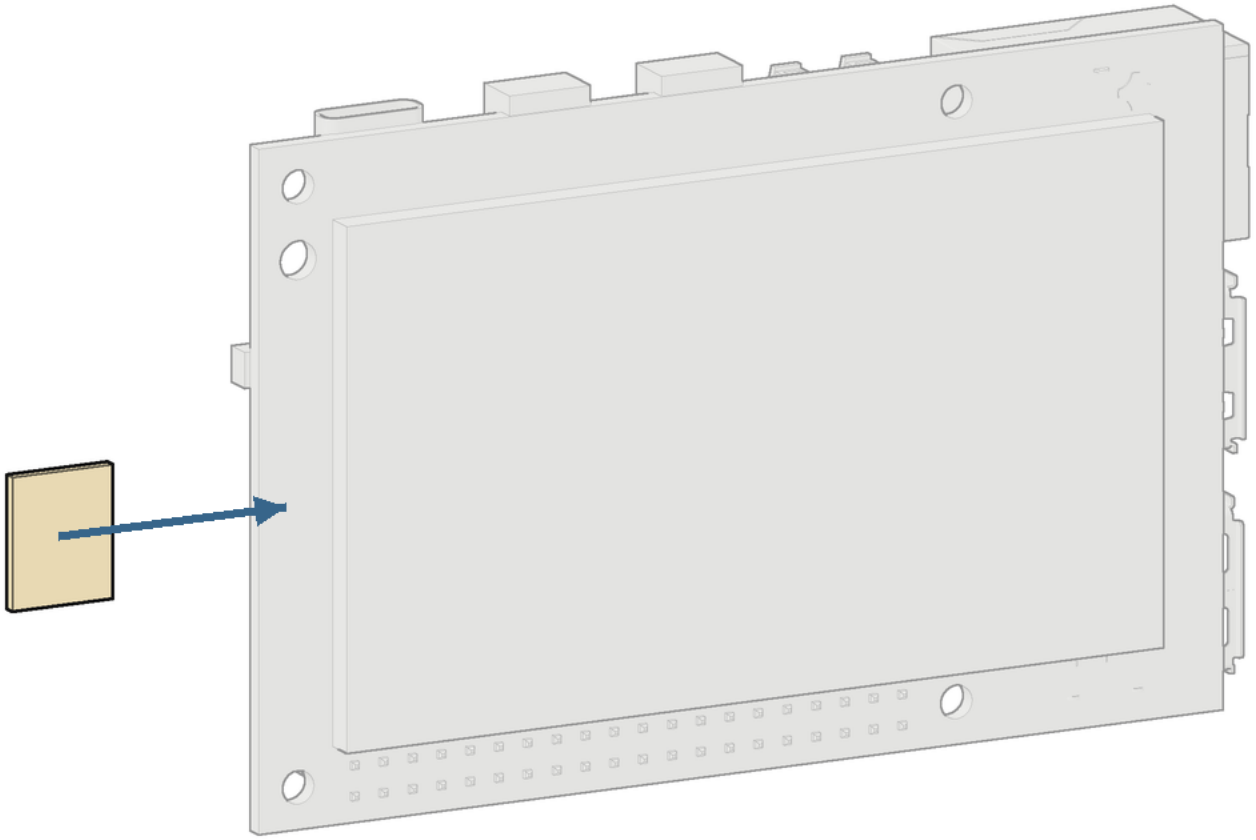
Keep pair mated in the camera. Bench separation only.

Amber cylinders reserve two finished splices, maximum diameter 3 x 10 mm. Cut lengths, solder overlaps, individual-core fanout and terminal joints still require measurement.

STOP Label FPS distinctly from PWR. These same-key connectors can be cross-connected. Stock and study axes are not cut instructions.

06d Fit the OS card on the bench

The microSD travels with the removable Pi stack. Closed-panel card access is blocked by the power switch.



Card before carrier

Prepare and verify the OS image on the bench.

Identify real contact orientation. Insert -X before the stack enters the body.

Leave the card with the stack during carrier insertion.

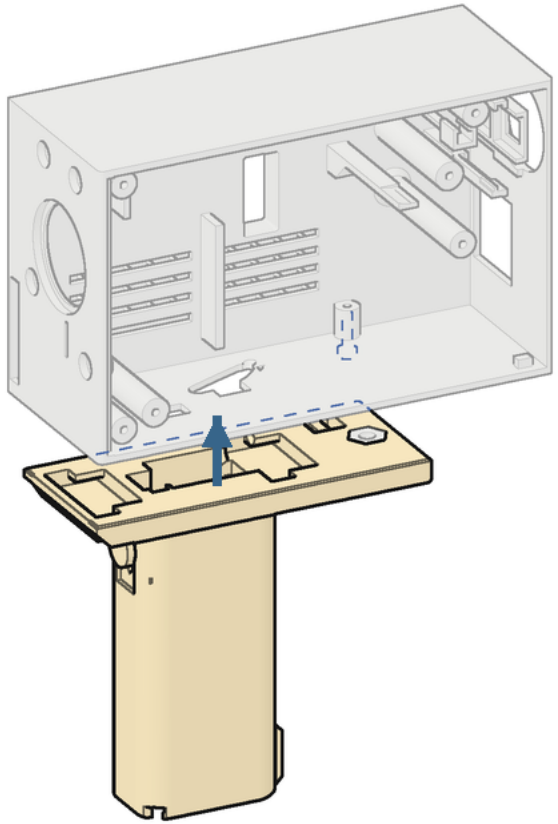
For later card service: isolate all power, remove the panel, then withdraw +X.

The panel-mounted J2 power switch occupies the card extraction route. The front slit alone does not establish service access.

STOP No closed-panel microSD service. Only a panel-off rigid path is checked; real contact orientation and handling remain open.

07 Join grip and tub

Battery stays out. Route the run lead while the floor passage is open.



Independent lock

Fit tripod nut and run button.

Engage keyhole tongues in their actual keyed direction.

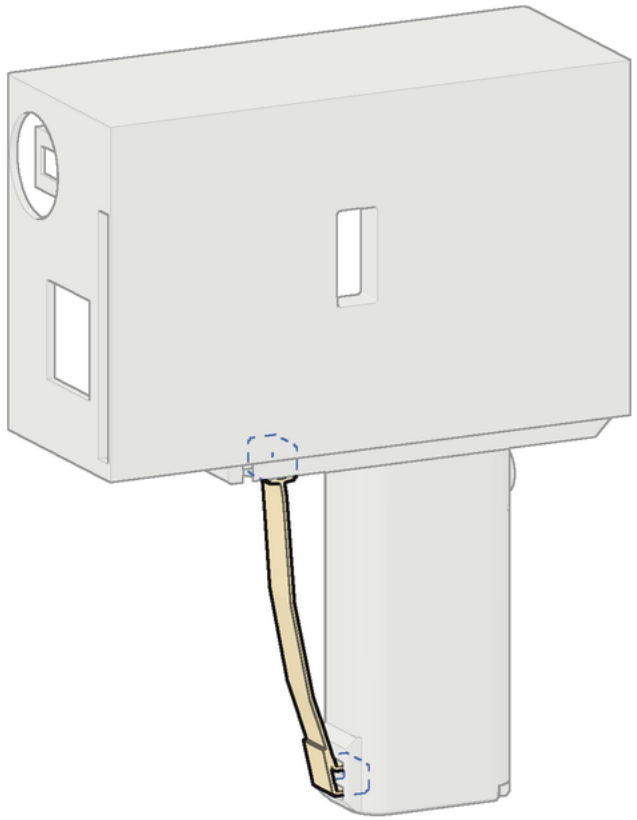
Use 1 x PT grip-lock screw.

Check strap and shifted lock clearances. Panel is not the grip lock.

STOP Arrow shows approach only. Keyhole engagement, driver access and actual lead passage need dry fitting.

07b Thread the strap after the grip lock

12 mm webbing through both upper and heel slots. Shown thickness 1.4 mm is a routing assumption.



Keep both returns visible

Thread after seating and locking the grip.

The lower return passes through both heel slots and inside the grip wall.

Keep the cap, battery travel and tripod seat clear.

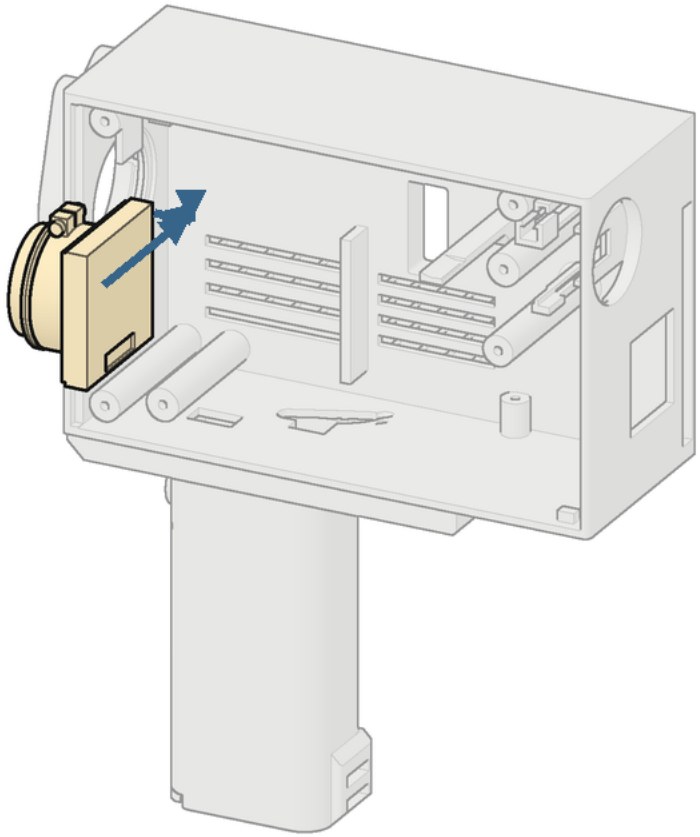
Buckle, closure overlap and retained tail still need real-part layout.

This is a full routing proxy, not a webbing cut pattern. Piecewise folds do not prove textile bend radius, hand clearance or physical threading.

STOP Neither anchor nor this routing is a rated safety tether. Qualify the actual strap, closure and both anchors before use.

08 Camera before the carrier

The board cannot be pushed through the installed vertical stack.



Three-part motion

1 Approach from +Y with rearward / upward offset.

2 Lower onto seating height.

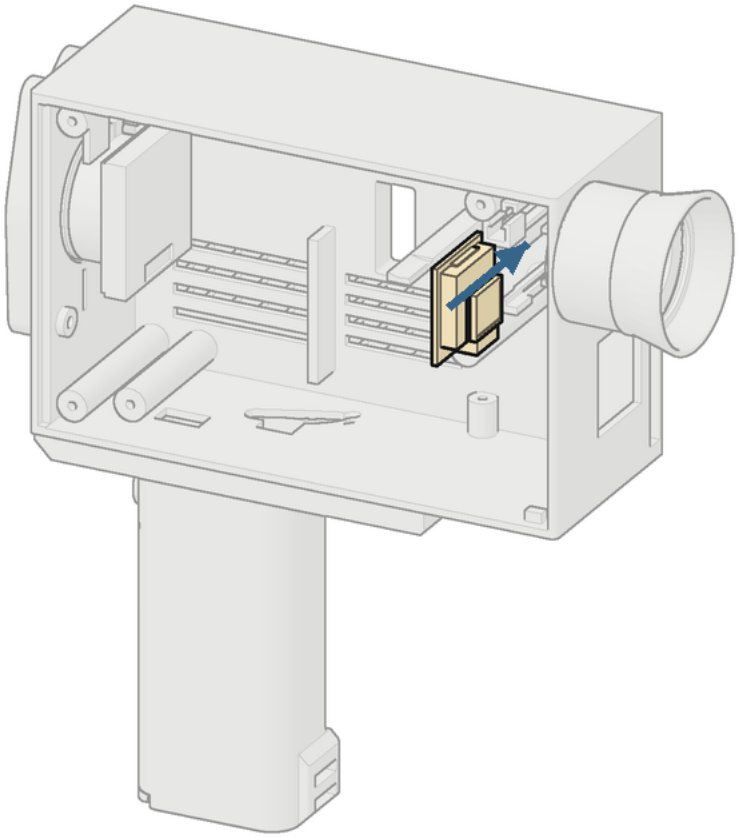
3 Move lensward to datum.

Fit collar with gauge and 3 anchor screws. Support optics throughout.

STOP Rigid camera path is sampled CAD only. Finger access, tolerances, optical support and torque catch remain open.

09 Seat the EVF module

Keep OLED foam, native flex and connector latches unstressed.



Keep the board secure

Slide module from +Y into its seats. Support the eyepiece.

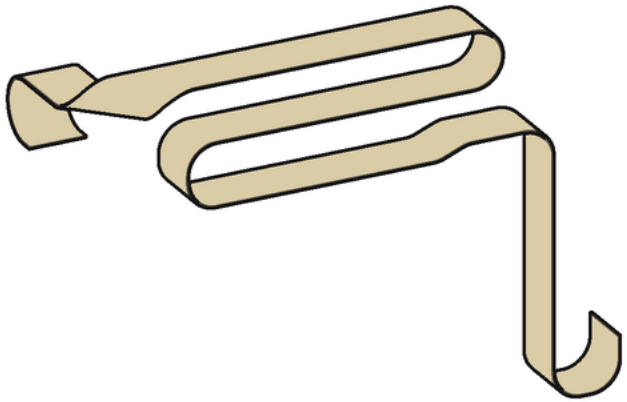
The final panel supplies clamping; do not let the open module fall.

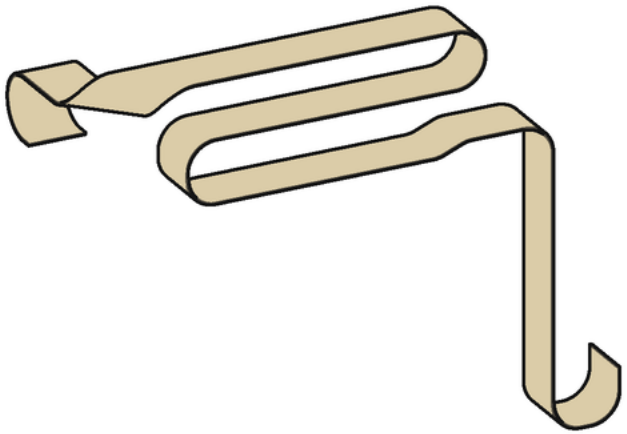
The straight HDMI head mates upward only after the board seats.

STOP Native flex contact face follows the actual EVF kit. Do not infer it from a CAD envelope.

10 Preform HDMI on the bench

3557 right-angle micro-HDMI at Pi | intact 300 mm 3562 ribbon | 3556 straight head at EVF





A full-length study

Each selected pose models 300 mm.

Keep Pi head mated during service.

Both-end-free pose rejected: power-wire conflict.

R3 / R5 are allowances, not manufacturer limits.

Installed

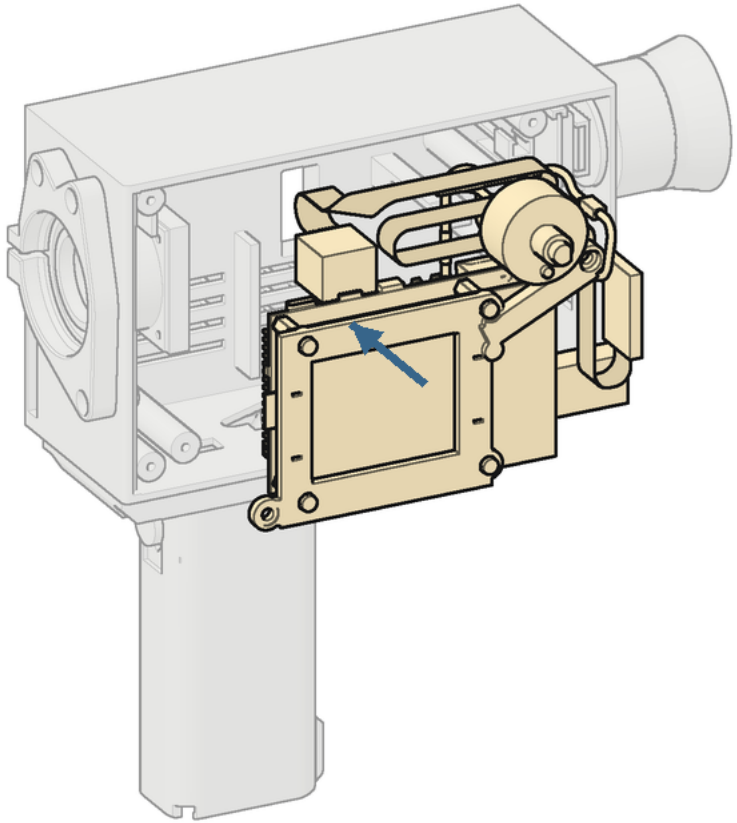
EVF end free / Pi still mated

The complete serpentine strip and both rigid heads must fit through every assembly and service pose.

STOP Full material clears checked solids; head/carry status is separate. Real bend, clocking, retention and EMI remain open.

11 Insert the stack with a free EVF end

Intended trial sequence: Pi HDMI adapter and OS microSD fitted first; recording drive absent.



Hold before proceeding

Support FPS bushing and stack separately; carry them together from +Y.

FPS pair and Pi HDMI stay mated. EVF HDMI end stays free.

Guide pigtail through floor opening.

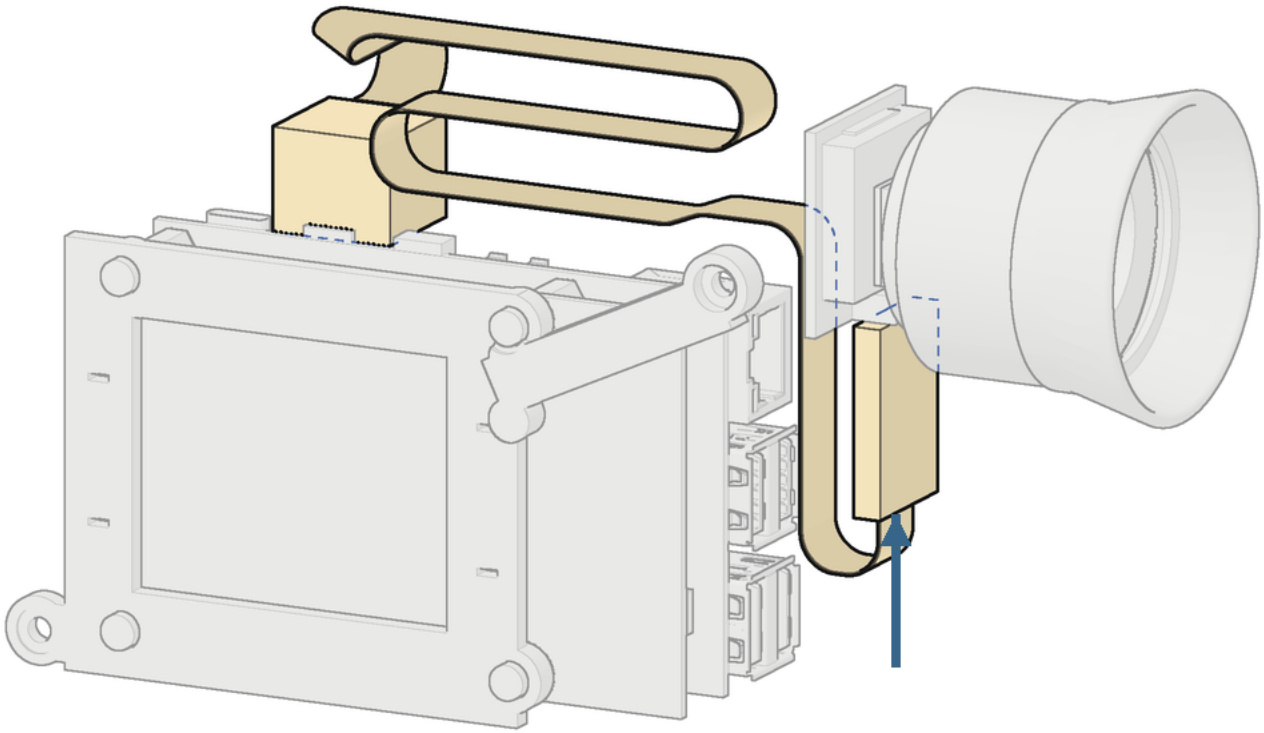
Seat carrier; 2 x PT screws. Do not pull leads to seat it.

CAD checked: full HDMI + carried heads at 2 mm poses through +Y70. FPS full mated harness shown separately; remaining loom motions are gated.

STOP Only use a sweep that includes every carried plug and the full ribbon. Bare-stack clearance is insufficient.

12 Mate HDMI without forcing the ribbon

Storage stays out to preserve access. Hold the adapter body, never pull the ribbon.



EVF head upward

Bring the straight 3556 head from below.

Confirm the keyed receptacle and actual contact face.

Mate fully by its body. Add qualified strain relief clear of the latch.

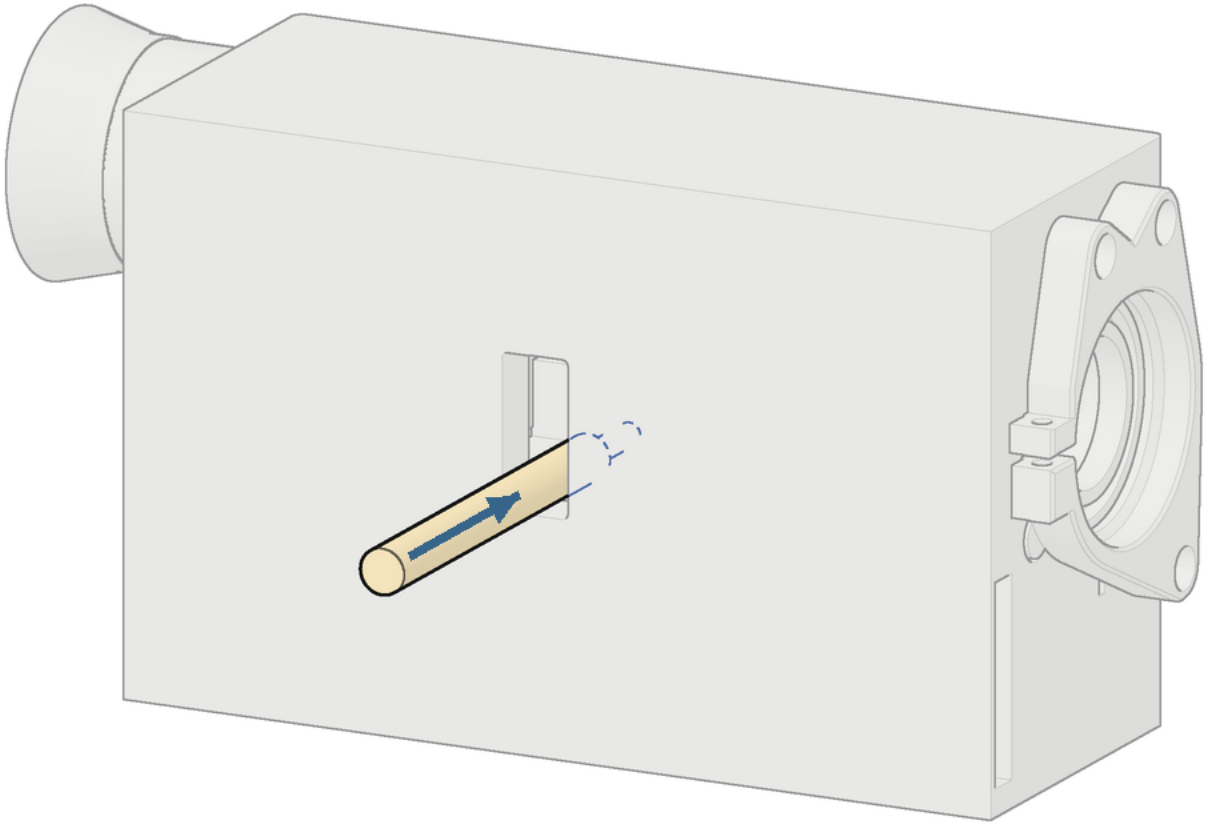
No twist to correct a wrong head.

Tub omitted for visibility. Adapter blocks are unmeasured catalog envelopes, not detailed keyed connectors.

STOP Upward head travel, fingers / pliers and simultaneous USB + HDMI access must all clear. Final pose is insufficient.

13 Prove camera connector access

12 x 27 mm R1 CSI opening = 323.14 mm2. All internal work remains unpowered.



Actual connector first

Do not cover the CSI opening.

CAM/DISP 1 -> official 22-to-15 FPC -> GS camera.

The inherited CSI proxy was wrongly oriented. Its old tool pass is rejected.

New reserve: 4 mm tip / 8 mm shaft at the connector ends.

Show real jaws, latch travel and the complete 200 mm camera FPC before attempting this step.

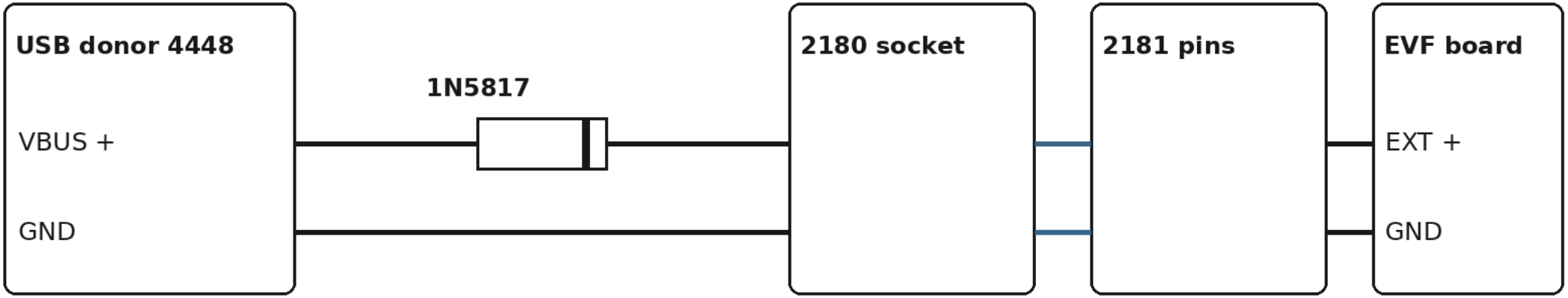
Rigid cylindrical reserve clears only; actual tool, cooler and full FPC remain unresolved.

Intake wall slots: 420 mm2 nominal. CSI remains a straight unsealed bypass; keep latch/tool access clear.

STOP OPERATIONAL HOLD: actual cooler outline, latch operation, inserted FPC and interacting flexible looms are unresolved.

13b Build the detachable EVF feed

Unpowered electrical schematic. Contact identity is traced by continuity, never colour alone.



Anode: USB side | Band / cathode: EVF side

USB side: 2 diode / positive joints + 1 ground splice. EVF side: 2 traced board joints.

Remove four donor jumper ends only after qualifying the length budget. Separately sleeve D+ and D-; insulate a shield only if present. Keep all joints off bends and strain-relieve both cable ends.

4448 length conflict: catalog 300 mm; supplier drawing 325 +/- 10 mm. Both Generic RCY-style tails: nominal 100 mm / 20 AWG. Final cuts, overlaps and actual plug / junction envelopes remain unmeasured.

STOP Isolate USB and battery. Meter polarity and diode direction. A diode does not guarantee the EVF voltage ceiling; G-W6 remains open.

14 Verify the battery and EVF power paths

Electrical intent only. Bench qualification is required before any full-camera power-up.

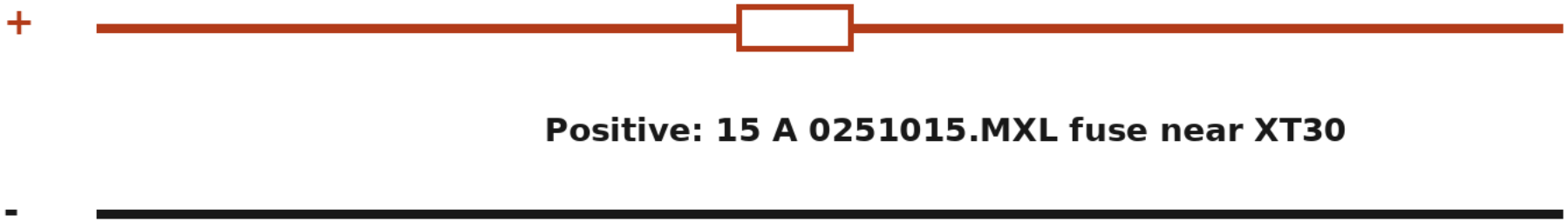
1S2P protected pack
60 mm 18 AWG lead

→

XT30 junction
Mate by housings

→

X1203 battery pads
Pogo feed -> Pi 5



16 AWG: buy 914.4 mm per colour. Both 270 mm path budgets include unresolved pad-root allowances. Positive also includes 20 mm rigid fuse / joint zone. Copper cuts / overlaps TBD.

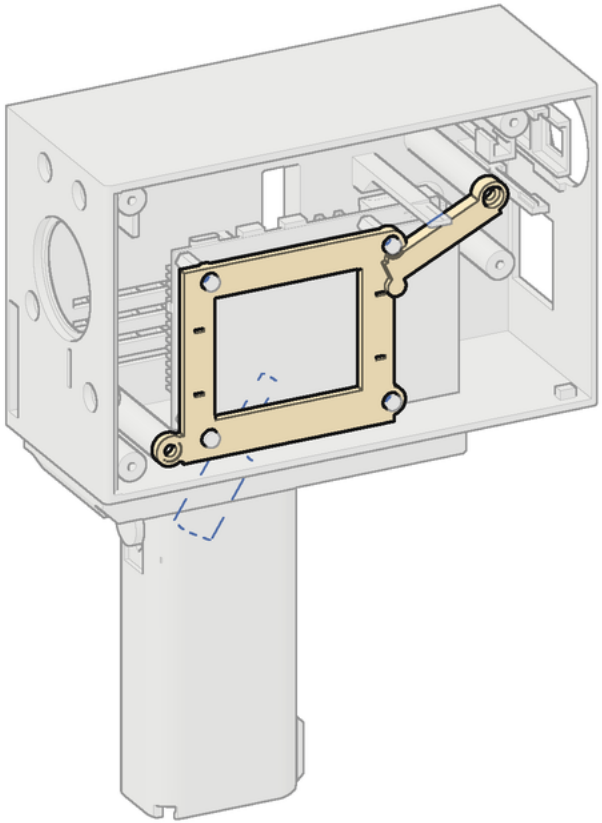
EVF USB positive -> qualified feed -> EXT+; USB ground -> EVF GND.

Inherited diode option: 1N5817 band / cathode toward EVF. It does not guarantee the required voltage ceiling. Regulator alternative is not adopted. Trace the actual EVF revision before soldering.

STOP Meter polarity / shorts. The 8.6 A / 0.15 V hot-drop gate remains open; the conservative 60 C copper model leaves only about 7 mV margin.

15 Prove stored slack and disconnection

Complete cable lengths, connector bodies, sleeves and tie heads must fit at the same time.



No hidden excess

Static wire capsules: clear printed solids; root joints omitted.

Keep fuse straight and supported. Keep joints off bends.

Four tie slots are provisions, not a tested retention scheme.

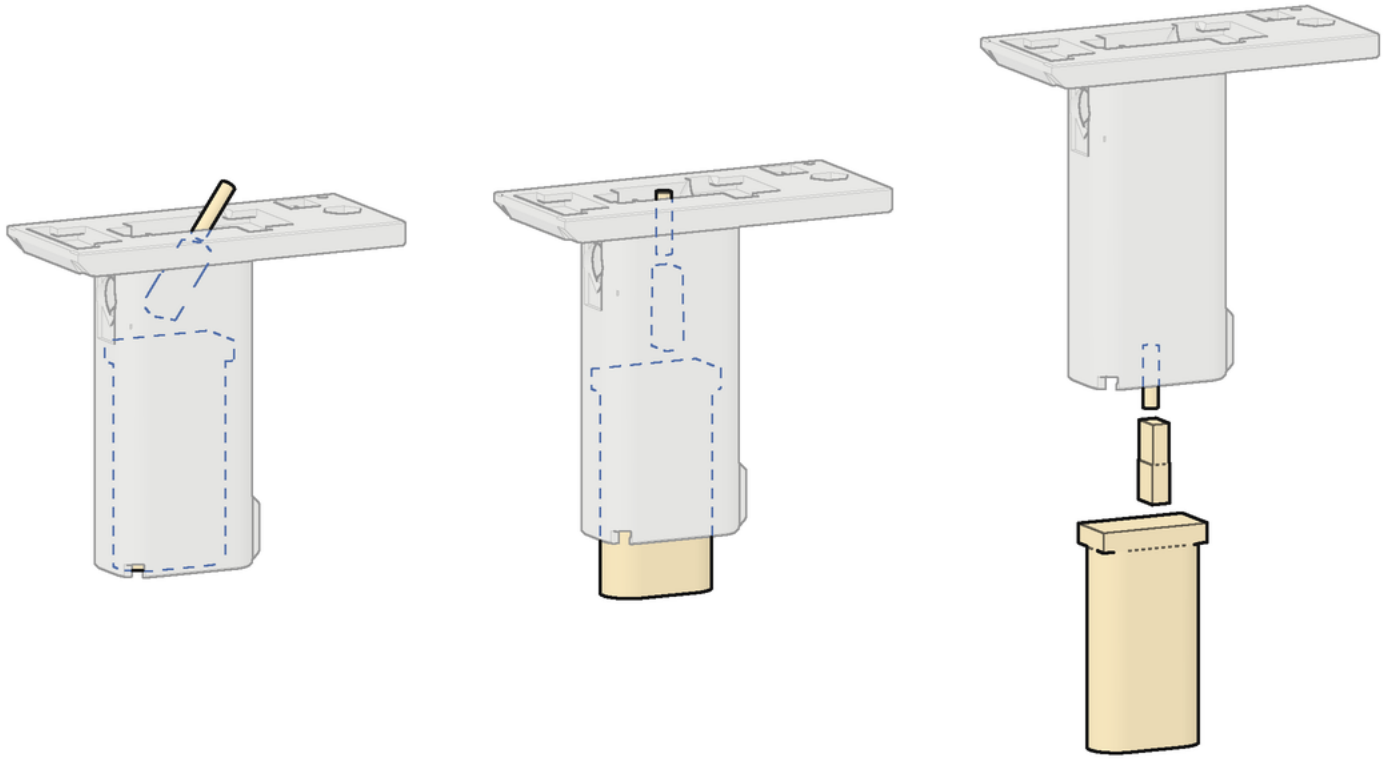
Loops must clear CSI tools, GPIO tails and UPS PCB.

Test pack extraction with retention unchanged.

STOP If XT30 cannot be parted before opening the body, the assembly is not safely serviceable. Redesign before closure.

15b Rigid connector withdrawal study

Body omitted for clarity; it stays closed during isolation. Full flexible-harness service is still blocked.



Stored **Unwind / lower** **Withdrawal limit**

What changed

CAD checked: 201 sampled poses clear the closed assembly, with pack moving and cap removed.

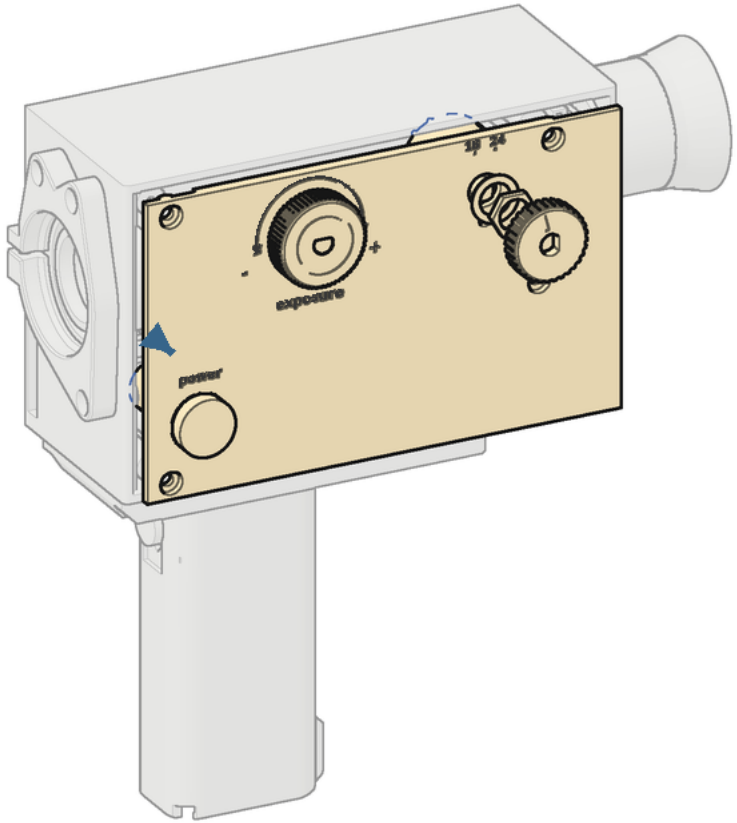
The pack moves with the junction.

This does not prove the complete conductor paths, joints and retained slack can deploy or return safely.

STOP Do not pull the pack through a blocked loom. Complete wires, pad joints, retention and pinch clearance remain STOPs.

16 Trial panel removal and refit

Opening shown. Reverse the support transfer for refit; keep the FPS pair mated.



Tip, then bushing

FPS stays on its rest.

Use gentle axial tip support for initial 8.5 mm.

Only 0.5 mm tip then remains; do not pinch it. Transfer support to accessible bushing.

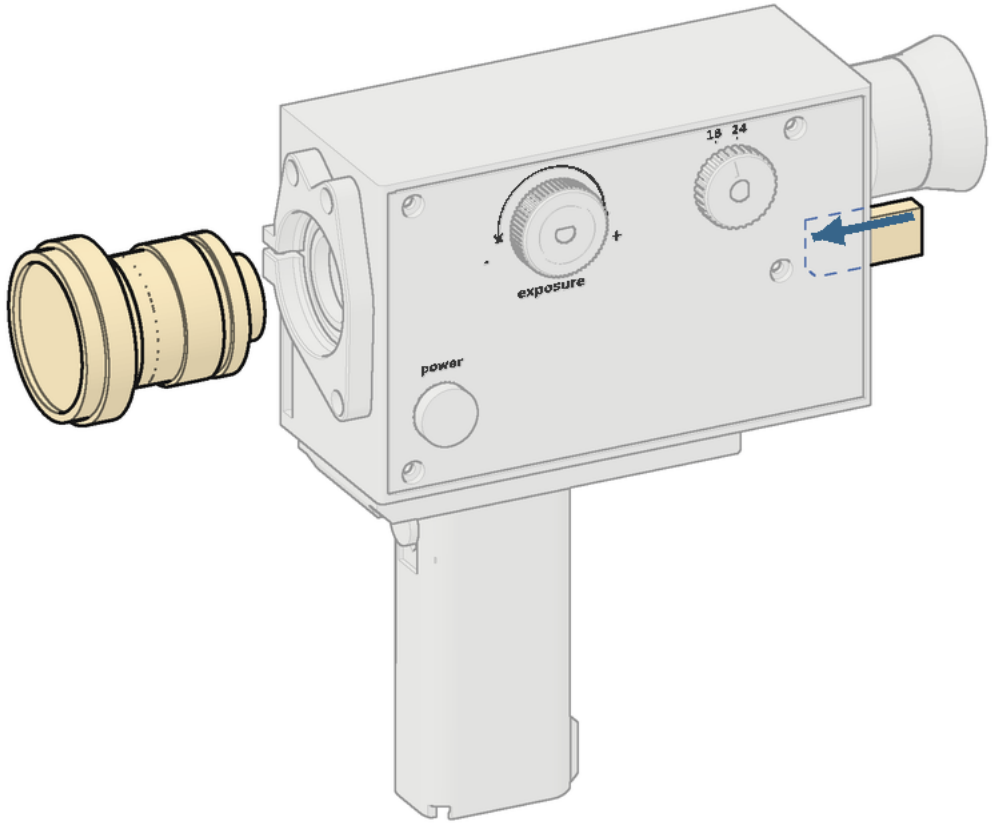
Refit: reverse support transfer. Preserve stop setting; washer, nut and knob go on last.

FPS pair stays mated.

STOP Never use screws to crush slack into the enclosure. Real shaft grip, stop-stack retention and whole-harness motion remain STOPs.

17 Recording drive and lens last

Fit selected compact Princeton drive with its protective cap removed. OS microSD is already installed.



Straight insertion

Drive travels +X into upper USB3.

OS microSD was fitted on the bench. Later card service requires panel removal.

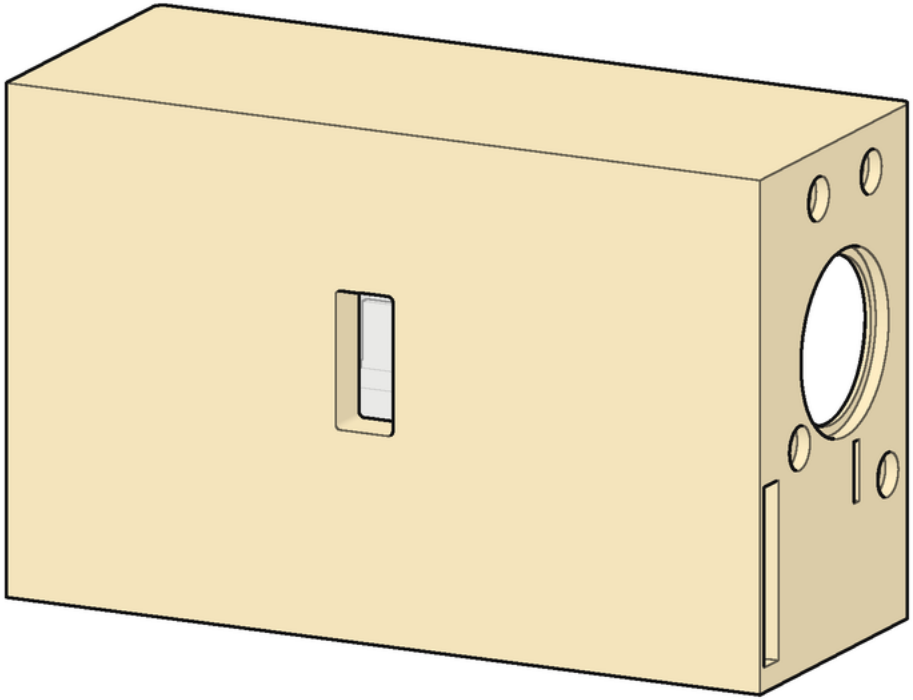
Support lens at collar; verify witness marks. Pinch screw is last optical operation.

Estimated storage-plug keepout omitted: it is not a physical inventory item or a proven mating shape.

STOP Published Princeton size includes cap. Real plug fit and warm sustained >=100 MB/s are not measured.

17v Keep ventilation clear during assembly

The outer bays conceal the inner slots. Their pressure loss, leakage, recirculation and cooling remain unmeasured.



Tub, stack and illustrative cooler. Panel and looms omitted for visibility; prove clearance with the complete final assembly.

Saved-STEP checks: 49 pass / 0 fail / 21 info. Smallest sampled hot-bay horizontal cut: 299.4 mm2. Mouth totals are not a global flow minimum.

BEFORE CLOSURE

Keep mouths, inner slots and baffles clear of the full FPC, EVF feed, power loop, tie heads and fan lead. All loom holds stay open.

CSI stays 12 x 27 mm R1 and uncovered. Its unsealed bypass means not all intake air must reach the cooler.

Test hand, strap and stand occlusion. Blocking the hot underside leaves 210 mm2 front opening for 510 mm2 wall slots. No blocked-vent operation is qualified.

THERMAL GATE

After electrical prerequisites pass, record ambient, closed-body load, fan speed/duty, component temperatures and throttling at normal hand positions. Compare with the open reference using the same hardware and conditions.

Measure the actual cooler first; its proxy/source conflict remains open.

STOP No airflow, fan operating-point, hand-occlusion or thermal pass. All 56 inherited physical gates, M1 and candidate gates stay open.

18 Close the loop before power

Physical checks and measured records decide whether this candidate can become a build.

- [ ] Pack disconnected during all internal work
- [ ] Connector identities, contact faces and continuity verified
- [ ] No trapped ribbon, fuse sleeve, splice, tie head or fan lead
- [ ] Panel flush; grip lock, strap and optical support secure
- [ ] Plenum mouths and wall slots clear; no trapped support; CSI open
- [ ] XT30 disconnect reachable with retention fitted
- [ ] Required bench gates passed before pack is connected last

Then qualify closed-body power, thermal, storage and recording performance, plus revised collar / strap loads. Nothing in this booklet closes a physical gate.

STOP 56 inherited gates + M1 + candidate gates remain open. No fabrication, safety or performance certification.

19 Service in dependency order

Always start with page 00 and physical battery disconnection.

Panel	FPS knob / nut / mounting washer off -> axial tip support + panel -> initial +Y8.5 -> bushing support -> detach QT / J2; keep FPS pair mated
Pi module	Drive out -X -> panel / EVF junctions detached -> release Pi FPC via qualified CSI access -> free pigtail / fuse -> stage carrier +Y with headers, supported FPS switch and full mated FPS harness; prove run reach
Camera	Support / remove lens -> panel off -> STOP until actual CSI latch release is proved -> Pi FPC free -> carrier removed -> camera FPC free -> reverse camera seating path
EVF	Panel off -> power junction and HDMI body unplugged -> flex protected -> module +Y
OS card	Panel off before in-body microSD withdrawal. Closed-panel route is blocked.
Refit	OS card on bench. Camera before carrier. Junctions before panel. Pack last.

STOP CSI tool clearance is not latch qualification. GPIO plugs stay carried; actual FPC release, run slack and wire withdrawal must pass.

20 Evidence and open checks

Source-linked review candidate. Vendor facts are not real-part measurements.

Snapshot

Mesh export: 2026-10-09T14:58:12Z. Generated 2026-10-09T15:45:15+00:00.

Geometry: compact-kowa_lm6hc meshes plus source-bound concealed-vent STEP checks; 300 mm HDMI snapshot. Wiring intent: candidate assembly/BOM and inherited WIRING. The separate receipt records exact source hashes.

What the CAD motions check

Rigid body motions are sampled, not continuous swept-volume proofs. Complete HDMI shapes and carried heads have a separate source-bound check. The source-registered CSI ends have a limited declared-tool check; real cooler, latch and full FPC access remain blocked. Other flexible leads, hands, tolerances and physical retention remain unqualified.

Primary sources checked 2026-10-09

Pi J2 and GPIO reference

Pi right-angle head 3557

EVF straight head 3556

Intact 30 cm ribbon 3562

Compact Princeton SSD

Lorlin CK rotary switch drawing

USB donor 4448 + supplier drawing

EVF source socket half 2180

EVF pin half 2181

Molex male assembly drawing

STOP No manufacturer bend limit, purchased cable measurement or production release is invented here.